

Chapter 10

Lecture 10: Frechet space. Hausdorff space. Continuity.

Definition 10.0.1. A topological space X is Hausdorff or T_2 if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \subseteq X$ such that $x \in U$, $y \in V$, and $U \cap V = \emptyset$.

Property 10.0.1. If X is a metric space, then X is Hausdorff.

Proof Let $d = d(x, y) > 0$. Take $U = B(x, \frac{d}{2})$ and $V = B(y, \frac{d}{2})$. Then $U \cap V = \emptyset$.

Property 10.0.2. For a Hausdorff space X , for every $x_0 \in X$, the singleton set $\{x_0\}$ is closed.

Let $x \in X$ with $x \neq x_0$. Since X is Hausdorff, there exist open sets $U, V \subseteq X$ such that $x_0 \in U$, $x \in V$, and $U \cap V = \emptyset$. Thus $V \subseteq X \setminus \{x_0\}$. Therefore, $X \setminus \{x_0\}$ is open, which means $\{x_0\}$ is closed. ■

Question If $\forall x_0 \in X$, $\{x_0\}$ is closed, is X Hausdorff?

Answer No. Consider the finite complement topology.

Definition 10.0.2. If every point $x_0 \in X$ is closed, then X is called a Frechet space or T_1 space.

Remark If X is Hausdorff(T_2), then X is T_1 . The converse is not true.

Property 10.0.3. X is T_1 if and only if $\forall x, y \in X, \exists U \ni x$ a open set such that $y \notin U$. That is, for all $x, y \in X$, there exists an open set containing x but not containing y .

Exercise Prove this property.

Exercise Let X satisfy the following: for $x, y \in X$ with $x \neq y$, either $\exists U \ni x$ open set such that $y \notin U$, or $\exists V \ni y$ open set such that $x \notin V$. Is it true that X is T_1 ?

Solution No. Consider $X = \{a, b\}$ with the topology $\mathcal{T} = \{\emptyset, X, \{a\}\}$.

Theorem 10.0.1. Let X be T_1 , and $A \subseteq X$. Then $x \in A'$ if and only if every open set $U \ni x$ contains infinitely many points of A .

Proof

($\Rightarrow$) Assume the contrary that only finitely many points are in the intersection. Suppose $x \in A'$. Then every open set $U \ni x$ contains a point of A different from x , i.e., $\exists y \in U \setminus \{x\} \cap A$. So we have that $U \cap A = \{y_1, y_2, \dots, y_m\}$. Since X is T_1 , then $\{y_i\}$ is closed for each $i = 1, 2, \dots, m$. Thus $V =: U \setminus \{y_1, y_2, \dots, y_m\}$ is open and contains x . Then $V \setminus \{x\} \cap A = \emptyset$, which contradicts the assumption that $x \in A'$.

($\Leftarrow$) Obvious.

Theorem 10.0.2. A sequence (x_n) in a Hausdorff space X converges to at most one point.

Proof Suppose a_1, a_2 are two distinct limits of the sequence (x_n) . Since X is Hausdorff, there exist open sets $U, V \subseteq X$ such that $a_1 \in U$, $a_2 \in V$, and $U \cap V = \emptyset$. Since a_1 is a limit of the sequence, there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $x_n \in U$. Similarly, since a_2 is a limit of the sequence, there exists $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$, $x_n \in V$. Let $N = \max\{N_1, N_2\}$. Then for all $n \geq N$, $x_n \in U$ and $x_n \in V$, which implies $x_n \in U \cap V$. This contradicts the fact that $U \cap V = \emptyset$. Therefore, the sequence (x_n) converges to at most one point. ■

Theorem 10.0.3. 1. $(X, <)$ with the order topology is Hausdorff.

2. If X is Hausdorff, then the set with subspace topology $Y \subseteq X$ is also Hausdorff.
3. If X_1, X_2 are Hausdorff, then the product space $X_1 \times X_2$ with the product topology is also Hausdorff.

Exercise Prove this theorem.

Definition 10.0.3. Let X, Y be topological spaces. A function $f : X \rightarrow Y$ is called continuous if for every open set $V \subseteq Y$, the preimage $f^{-1}(V)$ is an open set in X .

Remark If the topology on Y is given by the basis $\mathcal{B}$, then f is continuous if and only if for every basis element $B \in \mathcal{B}$, $f^{-1}(B)$ is open in X .

Example 10.0.1. Consider the identity map for topological spaces $\mathbb{R}$ with the usual topology and $\mathbb{R}_l$ with the lower limit topology. The identity map $id : \mathbb{R} \rightarrow \mathbb{R}_l$ is not continuous, but the inverse $id : \mathbb{R}_l \rightarrow \mathbb{R}$ is continuous. Note that $(a, b) = \bigcup_{n=1}^{\infty} [a + \frac{1}{n}, b)$.

Observation If $\mathcal{T}, \mathcal{T}'$ are topologies on X with $\mathcal{T} \subseteq \mathcal{T}'$, then the identity map $id : (X, \mathcal{T}') \rightarrow (X, \mathcal{T})$ is continuous and the map $id : (X, \mathcal{T}) \rightarrow (X, \mathcal{T}')$ is not continuous.

Theorem 10.0.4. The following are equivalent:

1. $f : X \rightarrow Y$ is continuous.
2. $\forall A \subseteq X, f(\bar{A}) \subseteq \overline{f(A)}$.
3. For every closed set $C \subseteq Y$, the preimage $f^{-1}(C)$ is closed in X .
4. $\forall x \in X$, for every open set $V \subseteq Y$ containing $f(x)$, there exists an open set $U \subseteq X$ containing x such that $f(U) \subseteq V$.

Proof

(1) $\Rightarrow$ (2) Let $f : X \rightarrow Y$ be continuous and $A \subseteq X$. Let $x \in \bar{A}$. We need to show that $f(x) \in \overline{f(A)}$. If $x \in A$, then $f(x) \in f(A) \subseteq \overline{f(A)}$. If $x \in \bar{A} \setminus A$, take $V \ni f(x)$ open in Y . Since f is continuous, $f^{-1}(V)$ is open in X and contains x . Thus $f^{-1}(V) \cap A \neq \emptyset$. Let $y \in f^{-1}(V) \cap A$. Then $f(y) \in V \cap f(A) \neq \emptyset$. Therefore, $f(x) \in \overline{f(A)}$.

(2) $\Rightarrow$ (3) Let $C \subseteq Y$ be closed. We need to show that $f^{-1}(C) := A$ is closed in X , i.e., $\bar{A} = A$. We have $f(A) = f(f^{-1}(C)) = C$. So if $x \in \bar{A}$, $f(x) \in \overline{f(A)} = \overline{C} = C$. So $x \in f^{-1}(C) = A$. Thus $\bar{A} \subseteq A$. The other direction is obvious.

(3) $\Rightarrow$ (1) Obvious (take the complement).

(1) $\Rightarrow$ (4) Let $f(x) \in V$. Take $x \in f^{-1}(V) =: U$ which is open in X . Then $f(U) \subseteq V$.

(4) $\Rightarrow$ (1) Let $V \subseteq Y$ be open. We need to show that $f^{-1}(V)$ is open in X . Let $x \in f^{-1}(V)$. Then $\exists x \in U_x \subseteq X$ such that $f(U_x) \subseteq V$. Take $U = \bigcup_{f(x) \in V} U_x$ which is open in X . So $f^{-1}(V) \subseteq U$. But $U \subseteq f^{-1}(V)$.

10.1 Constructing continuous functions

Theorem 10.1.1. Let X, Y, Z be topological spaces.

1. $f : X \rightarrow Y$ is a constant function, i.e., $f(X) = y$. Then f is continuous.
2. Let $A \subseteq X$ be a set with subspace topology. Then the embedding map $j : A \hookrightarrow X$ defined by $j(x) = x$ is continuous.
3. Let $f : X \rightarrow Y, g : Y \rightarrow Z$ be continuous, then $g \circ f : X \rightarrow Z$ is continuous.
4. Restriction of domain: Let $f : X \rightarrow Y$ be continuous and $A \subseteq X$, then $f|_A : A \rightarrow Y$ is continuous.
5. $f : X \rightarrow Y$ and $Z \subseteq Y$ with $f(X) \subseteq Z$. Then $f : X \rightarrow Z$ is continuous.
6. If $f : X \rightarrow Y, Y \subseteq Z$, then $f : X \rightarrow Z$ is continuous.
7. $f : X \rightarrow Y$ is continuous if $X = \bigcup U_\alpha$ (U_α is required to be open in X) and $f|_{U_\alpha} : U_\alpha \rightarrow Y$ is continuous.

Poof

1. Let $V \subseteq Y$ be open. If $y \in V$, then $f^{-1}(V) = X$ which is open. If $y \notin V$, then $f^{-1}(V) = \emptyset$ which is open.
2. Let $U \subseteq X$ be open. Then $j^{-1}(U) = U \cap A$ which is open in A .

3. $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W))$ which is open in X .
4. $(f|_A)^{-1}(V) = f^{-1}(V) \cap A$ which is open in A (Subspace topology).
5. Let W be open in Z . By subspace topology, $\exists V$ open in Y such that $W = V \cap Z$. Then $f^{-1}(W) = f^{-1}(V \cap Z) = f^{-1}(V)$ (because $f(X) \subseteq Z$) which is open in X .
6. Let $f : X \rightarrow Y$ be continuous and $Y \subseteq Z$ (Z has a topology $\mathcal{T}_Z$ and Y has the subspace topology $\mathcal{T}_Y = \{U \cap Y | U \in \mathcal{T}_Z\}$). Then we have for every open set $W \subseteq Z$, $f^{-1}(W) = f^{-1}(W \cap Y)$ which is open in X .
7. Let $X = \bigcup U_\alpha$ and U_α is open in X . And for each α , $f|_{U_\alpha} : U_\alpha \rightarrow Y$ is continuous.

Take $V \subseteq Y$ open. Then

$$f^{-1}(V) = f^{-1}(V) \cap X = f^{-1}(V) \cap \left(\bigcup U_\alpha \right) = \bigcup (f^{-1}(V) \cap U_\alpha) = \bigcup (f|_{U_\alpha})^{-1}(V) \quad (10.1)$$

Since $(f|_{U_\alpha})^{-1}$ is continuous, then $(f|_{U_\alpha})^{-1}(V)$ is open in U_α . And since U_α has the subspace topology and U_α is open in X , then $(f|_{U_\alpha})^{-1}(V)$ is open in X . Thus $f^{-1}(V)$ is open in X .

Theorem 10.1.2. Let $X = A \cup B$ where A and B are both open in X or both closed in X . And $f : A \rightarrow Y$, $g : B \rightarrow Y$ be continuous and $f(x) = g(x)$ for $x \in A \cap B$. Then define $h : X \rightarrow Y$ as

$$h(x) = \begin{cases} f(x) & \text{if } x \in A \\ g(x) & \text{if } x \in B \end{cases}$$

Then h is continuous.

Proof Let's assume that A, B are both open. Let $V \subseteq Y$ be open. Then

$$h^{-1}(V) = \{x \in X | h(x) \in V\} = \{x \in A | f(x) \in V\} \cup \{x \in B | g(x) \in V\} = f^{-1}(V) \cup g^{-1}(V) \quad (10.2)$$

Since f, g are continuous, by the subspace topology, $f^{-1}(V) = A \cap U_1$ where U_1 is open in X , and $g^{-1}(V) = B \cap U_2$ where U_2 is open in X . Thus

$$h^{-1}(V) = (A \cap U_1) \cup (B \cap U_2) \quad (10.3)$$

is open in X since A, B, U_1, U_2 are all open in X .

For the case where A, B are both closed, we have a similar argument. ■

Exercise Prove the above theorem.